# Parallel Lines: Extension

Name _____ Date _____ No calculator Version 1

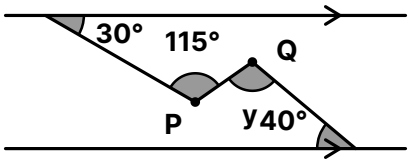
**KEY WORDS** parallel · alternate · co-interior · corresponding · proof · bisector · parallelogram · equation

## A The double zigzag ●●○ WORK IN YOUR BOOK

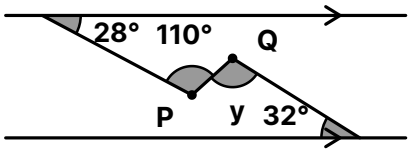
**Remember:** Draw a parallel line through each bend. Every angle at a bend splits into two alternate angles. Work from the angles you know to the ones you do not.

**1.** The two lines with arrows are parallel. Find the angle marked y. Show every step.

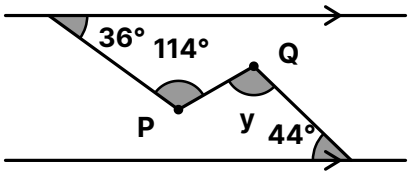
a)



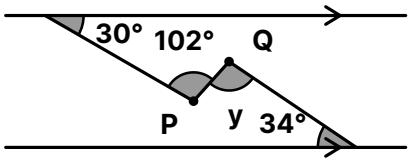

b)




c)




d)

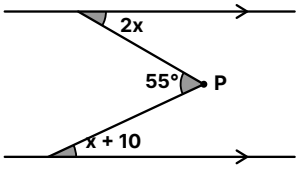



## B Equations from the figure ●●● WORK IN YOUR BOOK

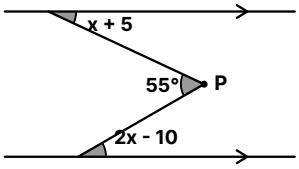
**Remember:** Decide first which rule links the marked angles (equal, add to 180°, or add to the third angle). Write that as an equation in $x$, solve it, then find every angle.

**2.** The two lines with arrows are parallel. Write an equation, solve it, and find each angle.

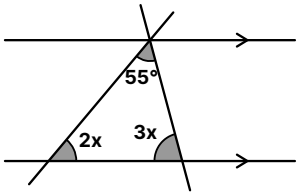
a)



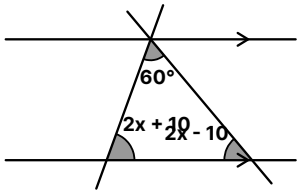

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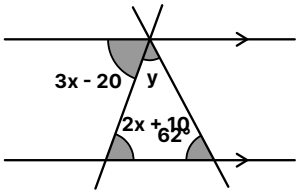

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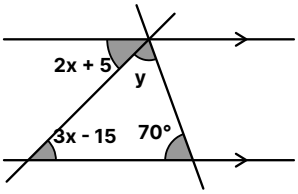

d)




e)




f)





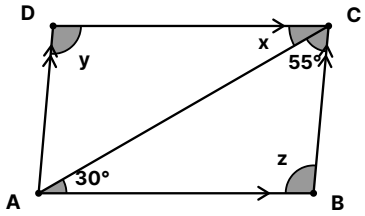

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## C Diagonals ●●● WORK IN YOUR BOOK

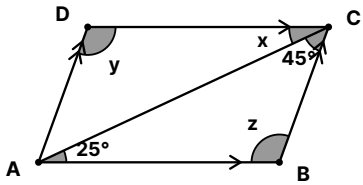
**Remember:** A diagonal is a line crossing two parallel sides, so it makes alternate angles with them. Then use the angle sum of the triangle it creates.

**3.** In each figure, AB is parallel to DC. Find the angles marked with letters. Give reasons.

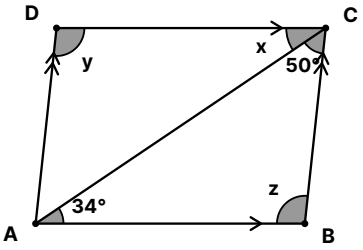
**a)**



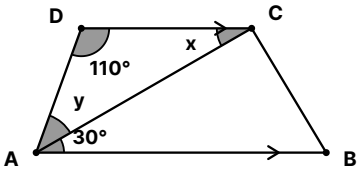

**b)**



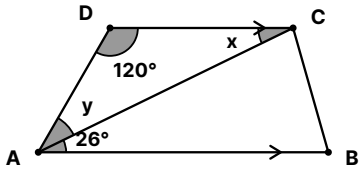

**c)**



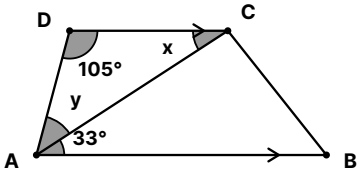

**d)**




**e)**



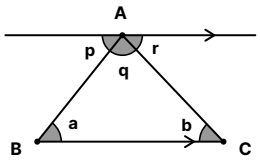

**f)**




## D Proofs ●●● WORK IN YOUR BOOK

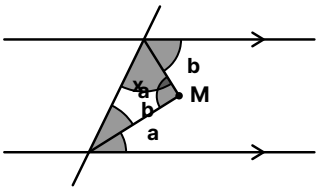
**Watch out:** A proof uses letters, not numbers, so it works for every triangle or every pair of lines. Give a reason for each line of the argument.

**4.** The line through A is parallel to BC. Use the figure to prove that the angles of a triangle add to 180°.




**a)** Write the proof.

**5.** The two lines with arrows are parallel. The co-interior angles at the two crossings are bisected (cut in half), and the bisectors meet at M.




**a)** Prove that the angle x at M is a right angle.

## ★ Extension FOR EARLY FINISHERS

**6.** ABCD is a parallelogram with the diagonal AC. Angle BAC is $40^\circ$ and angle ACB is $52^\circ$.

**a)** Without a figure, find all four angles of the parallelogram. Explain each step.

**b)** Explain why the diagonal AC can only bisect angle A when the parallelogram is a rhombus.


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# Parallel Lines: Extension

**Name** _____ **Date** _____ No calculator **Version 2**

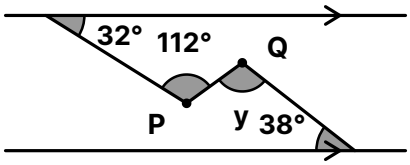
**KEY WORDS** parallel · alternate · co-interior · corresponding · proof · bisector · parallelogram · equation

## A The double zigzag ●●○ WORK IN YOUR BOOK

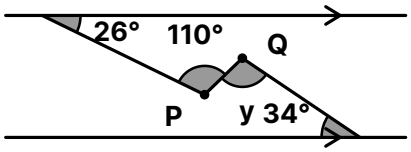
**Remember:** Draw a parallel line through each bend. Every angle at a bend splits into two alternate angles. Work from the angles you know to the ones you do not.

**1.** The two lines with arrows are parallel. Find the angle marked y. Show every step.

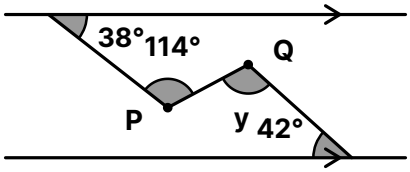
a)



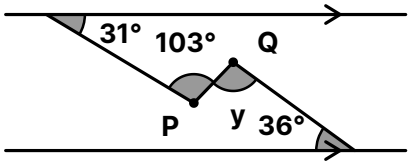

b)




c)




d)

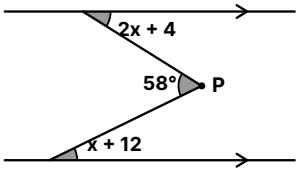



## B Equations from the figure ●●● WORK IN YOUR BOOK

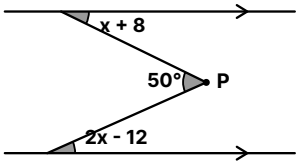
**Remember:** Decide first which rule links the marked angles (equal, add to 180°, or add to the third angle). Write that as an equation in $x$, solve it, then find every angle.

**2.** The two lines with arrows are parallel. Write an equation, solve it, and find each angle.

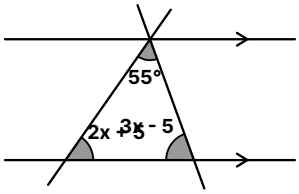
a)



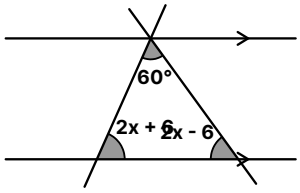

b)



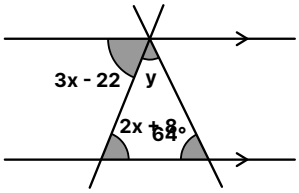

c)



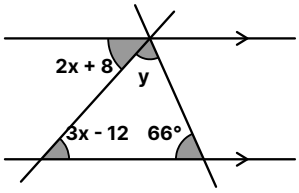

d)




e)




f)





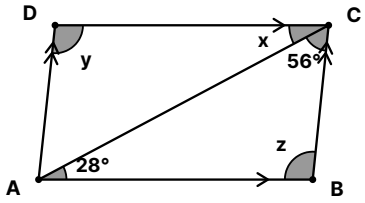

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## C Diagonals ●●● WORK IN YOUR BOOK

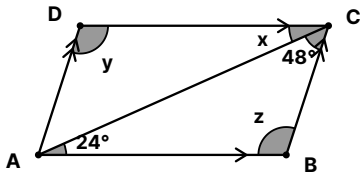
**Remember:** A diagonal is a line crossing two parallel sides, so it makes alternate angles with them. Then use the angle sum of the triangle it creates.

**3.** In each figure, AB is parallel to DC. Find the angles marked with letters. Give reasons.

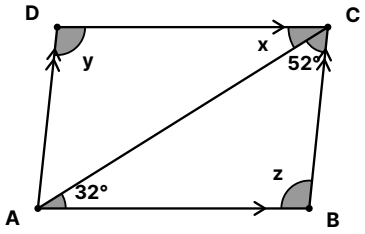
**a)**



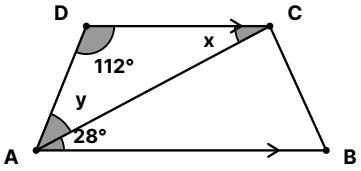

**b)**



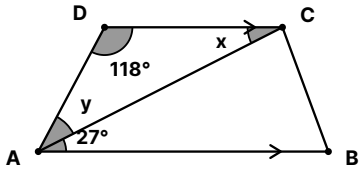

**c)**



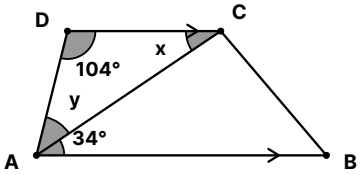

**d)**




**e)**



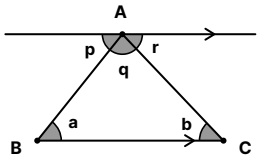

**f)**




## D Proofs ●●● WORK IN YOUR BOOK

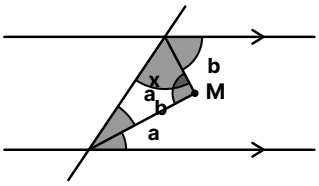
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**4.** The line through A is parallel to BC. Use the figure to prove that the angles of a triangle add to 180°.




**a)** Write the proof.

**5.** The two lines with arrows are parallel. The co-interior angles at the two crossings are bisected (cut in half), and the bisectors meet at M.




**a)** Prove that the angle x at M is a right angle.

## ★ Extension FOR EARLY FINISHERS

**6.** ABCD is a parallelogram with the diagonal AC. Angle BAC is $38^\circ$ and angle ACB is $54^\circ$.

**a)** Without a figure, find all four angles of the parallelogram. Explain each step.

**b)** Explain why the diagonal AC can only bisect angle A when the parallelogram is a rhombus.


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